



How We Tested

One of the most important factors that come into the picture, in a graphics card comparison, is the test environment. Even a slight variation in the configuration of the test machine is enough to ruin the results of the test. The test machine we used was an Intel Canterwood-based (D875 PBZ) computer, with a single Pentium 4 processor at 3.2 GHz and 512 MB DDR RAM (333 MHz), running Windows XP Professional. Before running all the tests, all the features of Windows XP that could potentially interfere with the normal operation of the tests, such as Windows Update, screen savers, power management, remote assistance, etc., were turned off. The operating system was re-installed after every test and brought to its initial state, so that it wouldn't cause any changes to the results. All the tests were run with V-sync and audio turned off, as V-sync synchronises the output of the graphics card with the refresh rate of the monitor, thereby giving frame rates that do not reflect the true capacity of the graphics card. All additional settings required for the tests were adjusted using the native drivers provided for the particular card being tested.

The tests for the graphic cards were extensive, and cover the entire range of features and specifications provided by the chipset. One very important thing to notice is that this test process has been revised for the DirectX 9 video cards, and the test process now includes two synthetic benchmarks, and one game for both the OpenGL and Direct 3D categories.

Features: Features of the card are assigned 10 per cent of the total score. All the bells and whistles such as TV-Out, S-Video, RCA, remote control and the software bundled with the video card are considered here.

Performance: The performance of the cards in the various tests made up for 90 per cent of the scores. We divided performance into three different categories—OpenGL, Direct3D and Image Quality settings.

OpenGL: In the OpenGL tests we used *Quake III* as the test game, as its game engine scales itself according to the provided resources.

It plays on both low-end, as well as high-end video cards, and so the scores logged show a marked difference in the fps. The synthetic OpenGL benchmark we used was VulpineGL that did a pretty good job at stressing the OpenGL subsystem of the graphics cards—the logged scores were almost one fourth of those given by the *Quake III* benchmark.

Direct3D: In the Direct3D tests, we used *Unreal Tournament 2003* as the game benchmark to test the card's performance when using DirectX 8.1. We used four of the standard benchmarks provided with *UT2003*—*BotMatch*, *Antalus and Flyby Antalus*, *BotMatch Asbestos and Flyby Asbestos*. The *GunMetal* benchmark is a pure DirectX 9 benchmark, which uses a real game engine to give the fps scores of the video card.

3D Mark 2003 was used as a synthetic benchmark to test the video cards, because it contains benchmarks to test the cards' Pixel and Vertex Shaders, and also subjects them to various other tests.

Image quality: We also tested the image quality of the screenshots taken during game play, since this gives an explicit idea of the quality of rendering supplied by the card. The image quality tests comprised of two in-game screenshots, taken while playing *UT 2003*, with all the eye-candy turned on. This helped to test the anti-aliasing and anisotropic filtering features of the video card.

The performance tests were given precedence over the features because better features obviously reflected better performance scores. Peripheral features such as Video-In, Video-Out were given additional points, thereby eliminating the tendency of certain vendors to obtain better scores by resorting to 'specmanship'—the art of bombarding a user with such impressive specifications, that he believes it's a better product.

All in all, these tests benchmark each and every aspect of the graphics cards, including the GPU and the memory bandwidth efficiency available to the graphics card.

chip, and features such as anisotropic filtering and anti-aliasing were nice and solid. Suddenly nVidia had some serious competition from a player they had left to die. And as time went by, ATi just got bigger and better.

Chipsets in the pipeline

OpenGL has a rendering pipeline that directs different elements. Both nVidia and ATi have announced their latest graphics cards—the nVidia NV40 and the ATi R420. The NV40 is rumoured to be manufactured by IBM as opposed to nVidia's long time fabrication partner Taiwan Semiconductor Manufacturing Corporation (TSMC). The NV40 has a 0.13 micron-based graphics core, just like the latest NV35 chipsets found on GFX 5900 graphics cards, but this is where the similarity ends. The NV40 is said to have almost twice the speed of NV35, and is based on a completely new architecture. We can only hope that this new technology does not turn into a huge embarrassment—as the NV30 (GFX 5800) did. ATi also claims that the R420 chipset-based graphic cards will be twice as fast as the Radeon 9800 Pro. The R420 will most probably be built on the 0.15 micron-based technology currently used for fabrication at TSMC.

The early Radeon cards such as the 7500 and its scaled down version, the 7500 LE, were targeted to match up with the GeForce3 and GeForce2 MX versions from nVidia. But they fell short because of poor drivers and some other major hitches. However, ATi did enjoy some success with this product line. Then it went back to the drawing board, and came out with the 8500—which targeted the GeForce4—and its souped-up version, the Radeon 9000.

With the current crop of DirectX 9 cards, ATi has come out with the Radeon 9700, and its successor, the 9800. We dug deep to see what lies under the hood of these cards.

Analysis

There is no such thing as 'the best 3D accelerator card,' since every user has a different set of requirements and what is 'best' for one, might not even be close to the expectations of another. Realising this, we broke up the graphics card market into three segments—the low end, the mid-range and the high-end cards. The low-end segment is completely dominated by the nVidia-based cards, because ATi does not offer any cards in this category. The fight, therefore, was between individual graphics cards based on the same chipset, with the results being fairly interesting. However, in the other segments, the fight was much more interesting, since both vendors had offerings.